

TASK 5

Capstone Integration & Portfolio Finalization

Sales Performance Intelligence • Final Portfolio Document

Project ApexPlanet Sales Performance Analysis	Tools Python • Pandas • Matplotlib • Power BI • Excel • Power Query • DAX
Portfolio purpose Consolidate Tasks 1–4 into one professional, presentation-ready project.	Final evidence Interactive Power BI dashboard + statistical validation + business recommendations

Portfolio Statement

This capstone brings the complete ApexPlanet sales analytics workflow together: data preparation, exploratory analysis, business-question analysis, interactive dashboarding, statistical validation, and decision-oriented recommendations.

Prepared for final internship submission

1. Master Portfolio Overview

The portfolio is organized as a single reference point for the complete internship journey.

Component	What it contains	Portfolio status
Task 1 — Data Preparation	Dataset inspection, missing-value checks, duplicate handling, validation and cleaned data preparation.	Completed
Task 2 — Exploratory Data Analysis	Univariate analysis, business questions, multivariate analysis, correlation and KPI exploration.	Completed
Task 3 — Power BI Dashboard	KPI cards, sales trend, category/product/city analysis, age and gender analysis, and slicers.	Completed
Task 4 — Data Storytelling & Validation	Business story, visual evidence, Welch two-sample t-test and action-oriented recommendations.	Completed
Task 5 — Integration	Master portfolio, professional reflection, repository structure, walkthrough plan and LinkedIn-ready summary.	This document

Recommended GitHub master-repository structure

Use a single repository such as ChinniSnehitha-DataAnalytics-Internship-Portfolio (or your preferred GitHub name).

- README.md — project homepage, internship overview, tools, findings and links.
- Task-1-Data-Preparation/ — notebook and cleaned-data documentation.
- Task-2-EDA/ — EDA notebook, charts and business-question analysis.
- Task-3-PowerBI-Dashboard/ — PBIX/report assets and dashboard screenshot.
- Task-4-Storytelling-Statistical-Validation/ — presentation, notebook and test results.
- Task-5-Portfolio/ — this final portfolio PDF and final reflection.
- assets/ — screenshots, supporting visuals and presentation images.

Links to add after GitHub upload

Master Repository: [PASTE GITHUB URL]

Task 1: [PASTE URL] Task 2: [PASTE URL]

Task 3 Dashboard: [PASTE URL] Task 4 Presentation: [PASTE URL]

2. Internship Work Consolidated

A compact record of the end-to-end workflow.

Task 1 — Data Preparation

Inspected the sales dataset, checked structure and data types, reviewed missing values and duplicates, standardized text fields, validated age/quantity/price ranges, and recalculated Total_Sales from Quantity × Unit_Price.

Task 2 — Exploratory Data Analysis

Used Python/Pandas and Matplotlib for distributions and category-level exploration. Business questions covered top product, city, category, gender and age group, followed by multivariate relationships and correlation analysis.

Task 3 — Interactive Power BI Dashboard

Built a sales-performance dashboard with five KPI cards, monthly trend analysis, category and product charts, city and age-group views, gender analysis, and interactive City, Category, Gender and Age Group slicers.

Task 4 — Data Storytelling & Statistical Validation

Converted descriptive analysis into a business story and tested whether male and female customers had different average transaction sales using an independent two-sample Welch t-test.

Core tools and skills demonstrated

- Python: Pandas for data preparation and analysis; Matplotlib for visual exploration.
- Data quality: missing-value handling, duplicate checks, validation rules and derived sales calculation.
- Business analytics: KPI definition, segmentation, trend analysis and business-question framing.
- Power BI: KPI cards, charts, slicers and interactive dashboard design.
- Statistics: Welch independent two-sample t-test, p-value interpretation and confidence interval reporting.
- Communication: data storytelling, recommendations, portfolio documentation and presentation planning.

3. Final Dashboard Evidence

The dashboard is the interactive evidence layer for the sales analysis.

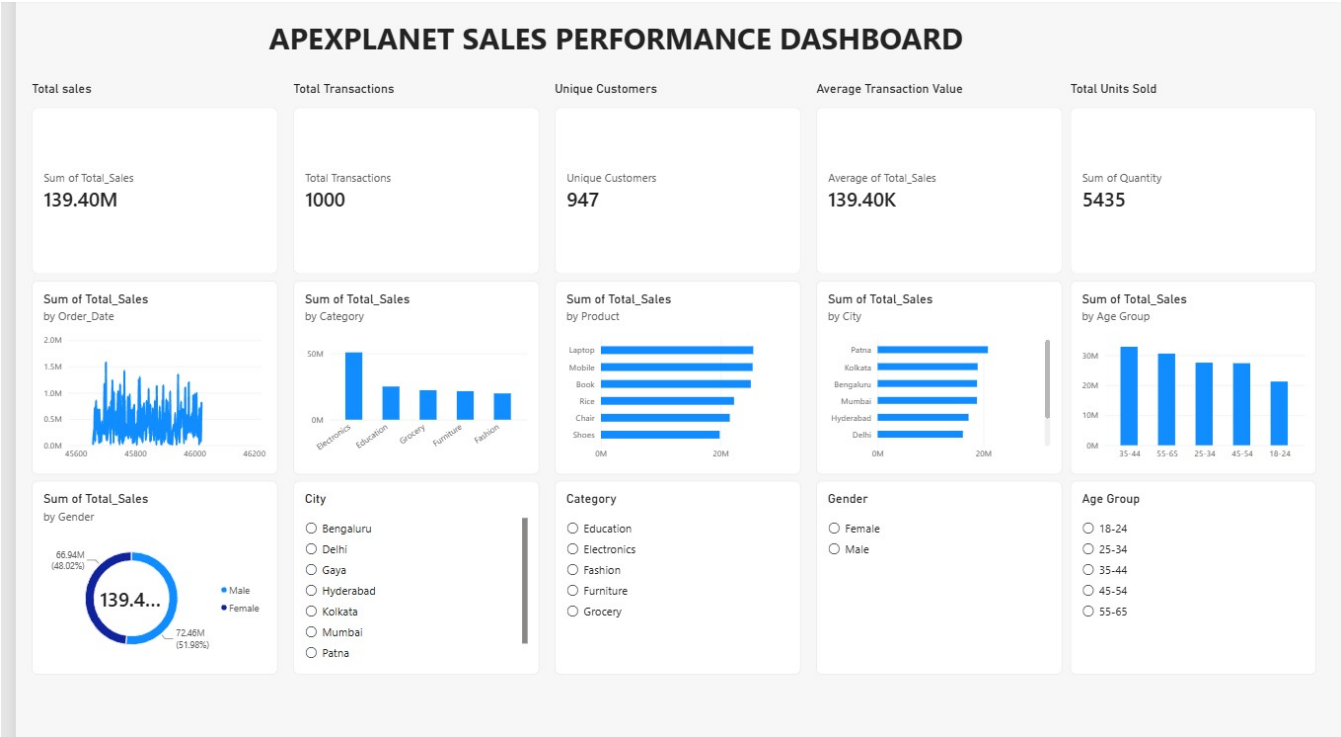


Figure 1. ApexPlanet Sales Performance Dashboard

Displayed KPI snapshot

KPI	Displayed value	Meaning
Total Sales	₹139.40M	Revenue represented in the dataset
Total Transactions	1,000	Sales records analyzed
Unique Customers	947	Distinct customers represented
Average Transaction Value	₹139.40K	Average sales per transaction record
Total Units Sold	5,435	Total quantity sold

4. Key Findings & Statistical Validation

Descriptive evidence and inferential evidence are kept separate.

Business findings

- Total sales are approximately ₹13.94 Cr across 1,000 sales records.
- Electronics is the leading displayed category at approximately ₹5.08 Cr.
- Laptop is the highest-revenue product at approximately ₹2.54 Cr.
- Patna is the highest-performing displayed city at approximately ₹2.08 Cr.
- The 35–44 age group contributes the highest sales in the displayed age-group analysis.
- The dashboard shows a near-even gender sales split: Female 51.98% and Male 48.02%.

Hypothesis test

Business question: Do male and female customers have different average transaction sales?

H_0 : Mean transaction sales are equal for male and female customers.

H_1 : Mean transaction sales are different.

Method: Independent two-sample Welch t-test, two-sided, $\alpha = 0.05$.

Statistic	Result
Male transactions	n = 511; mean = ₹141,807.34
Female transactions	n = 489; mean = ₹136,883.21
Welch t-statistic	0.683
Degrees of freedom	≈ 998.0
p-value	0.495
95% CI, Male – Female	₹−9,231.58 to ₹19,079.85

Statistical interpretation

At the 5% significance level, the test result is fail to reject H_0 ($p = 0.495$). In this dataset, the observed difference in average transaction sales between male and female customers is not statistically significant. The Task 4 report therefore advises against treating gender alone as a strong driver of average transaction value.

5. Business Recommendations & Professional Reflection

Decision-oriented recommendations

- Prioritize inventory and promotional planning around high-performing categories and products.
- Investigate weaker months for seasonal, pricing or promotional factors.
- Use age group, city, category and purchasing behavior together for segmentation.
- Develop loyalty and retention initiatives to increase repeat purchases.
- Monitor city-level performance to optimize regional marketing and inventory.
- Validate transaction identifiers before using operational KPIs.

Professional reflection

This internship project helped me understand how a raw business dataset can be converted into a complete analytics solution. I progressed from data inspection and cleaning to exploratory analysis, business-question formulation, dashboard development and statistical validation. The project also strengthened my ability to communicate findings rather than only produce charts. A key learning was the importance of checking data quality before defining KPIs—for example, repeated Order_ID values required careful interpretation of the transaction metric. The final workflow combines technical analysis with business context and provides a foundation for future projects in data analytics and business intelligence.

What I can demonstrate from this project

Technical: Python/Pandas, Matplotlib, Excel, Power Query, Power BI, DAX concepts and statistical testing.

Analytical: data cleaning, segmentation, KPI design, trend analysis, correlation and hypothesis testing.

Communication: dashboard storytelling, business recommendations, portfolio documentation and presentation walkthrough.

6. Final Walkthrough & LinkedIn-Ready Summary

3-5 minute portfolio walkthrough script

- 0:00-0:30 — Introduction: Introduce the ApexPlanet internship project and the objective of turning sales data into actionable business insights.
- 0:30-1:10 — Data preparation: Explain that the dataset was inspected, cleaned and validated before analysis, including duplicate and missing-value checks.
- 1:10-1:50 — EDA: Show the main distributions and business questions: product, city, category, gender and age-group sales.
- 1:50-2:40 — Power BI dashboard: Walk through the five KPI cards, monthly trend, category/product/city charts and the interactive slicers.
- 2:40-3:20 — Statistical validation: Explain the male-vs-female transaction-sales question, Welch t-test, p-value 0.495 and the interpretation.
- 3:20-4:00 — Recommendations: Discuss product/category focus, city optimization, segmentation and customer retention.
- 4:00-4:30 — Reflection: Summarize the skills gained and close with the GitHub portfolio link.

LinkedIn post draft

Copy-ready post

I'm happy to share my completed ApexPlanet Data Analytics Internship portfolio! 📄

Across the internship, I worked through an end-to-end sales analytics workflow—from data preparation and exploratory analysis to an interactive Power BI dashboard and statistical validation. The final project highlights sales KPIs, product/category performance, city and customer segmentation, monthly trends, and a Welch t-test examining gender differences in average transaction sales.

Key tools: Python, Pandas, Matplotlib, Excel, Power BI, Power Query and DAX.

This experience strengthened my data-cleaning, visualization, statistical-analysis and business-storytelling skills. I'm looking forward to applying these skills to more real-world analytics projects.

Portfolio: [PASTE MASTER GITHUB URL]

Dashboard: [PASTE DASHBOARD URL]

Final submission checklist

- ☐ Upload master GitHub repository.
- ☐ Add links to Tasks 1-4 and this Task 5 PDF.
- ☐ Upload the Power BI/report files and dashboard screenshot.
- ☐ Add the Task 4 presentation and statistical-validation notebook.
- ☐ Record the 3-5 minute portfolio walkthrough.
- ☐ Add GitHub and dashboard/demo links to the LinkedIn post.

7. Final Portfolio Source Record

Files used to consolidate the internship work.

Source	Role in final portfolio
APPEX_TASK_1 (1).ipynb	Task 1 data inspection and cleaning workflow.
APPEX_TASK_2 (1).ipynb	Task 2 EDA, business questions, multivariate analysis and KPI exploration.
ApexPlanet_Task3_Final_Document (1).docx	Task 3 dashboard documentation, KPI values, findings and recommendations.
APPEX_TASK_4.ipynb	Task 4 Welch t-test calculations and statistical decision.
ApexPlanet_Task4_Data_Storytelling_Statistical_Validation.pptx	Task 4 business story, visual evidence, statistical results and call to action.
ApexPlanet_DataAnalytics_Dataset_Cleaned.xlsx	Cleaned dataset used for the analysis.
ApexPlanet dashboard screenshot	Visual evidence of the completed Power BI dashboard.

Submission note

This PDF is designed as the final Task 5 portfolio/finalization document. Before submission, replace the bracketed GitHub/dashboard placeholders with the actual URLs and add the recorded walkthrough link if your internship portal requires one.

End of Portfolio